

- 6 (a) period = 15 ms C1
frequency = $1 / \text{period} = 1 / 0.015 = 67 \text{ Hz}$ A1
- (b) r.m.s current C1
 $= 0.75 / \sqrt{2}$ C1
 $= 0.53 \text{ A}$ A1
- (c) energy C1
 $= I_{\text{rms}}^2 R t = 0.53^2 \times 450 \times 30 \times 10^{-3}$ C1
 $= 3.8 \text{ J}$ A1
- (d) instantaneous power P dissipated in resistor is $P = I^2 R$ where I is instantaneous current and R is resistance of resistor B1
current still flows in resistor in opposite directions during half-cycles, so heating resistor B1
- OR
- average power $\langle P \rangle$ dissipated in resistor is $\langle P \rangle = I_{\text{rms}}^2 R$ where I_{rms} is root-mean-squared current and R is resistance of resistor (B1)
- I_{rms} not zero, so heating resistor (B1)

Section B

7 (a) progressive: energy transferred along direction of propagation of wave